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United States Patent	12407947
Kind Code	B2
Date of Patent	September 02, 2025
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Photoelectric conversion apparatus and equipment

Abstract

A photoelectric conversion apparatus included pixel circuits, signal lines, current sources, a switching circuit configured to switch a connection state among the plurality of signal lines and the plurality of current sources, and a control circuit configured to control the connection state of the switching circuit. The switching circuit can implement a first connection state in which a first signal line and a second signal line are insulated from each other, a first current source is electrically connected to the first signal line, and a second current source is electrically connected to the second signal line, and a second connection state in which at least the first current source is electrically connected to at least the second signal line. The control circuit selects the first and second connection states in first and second operation modes, respectively.

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Appl. No.:	17/963289
Filed:	October 11, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230171514 A1	Jun. 01, 2023

Foreign Application Priority Data

JP	2021-185152	Nov. 12, 2021
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Publication Classification

Int. Cl.: H04N25/42 (20230101); H04N25/709 (20230101); H04N25/767 (20230101); H10F39/00 (20250101)

U.S. Cl.:

CPC H04N25/42 (20230101); H04N25/709 (20230101); H04N25/767 (20230101); H10F39/8037 (20250101);

Field of Classification Search

CPC: H01L (27/14627); H01L (27/14636); H01L (27/14634); H01L (27/14621); H04N (5/378); H04N (25/78); H04N (25/767); H04N (25/709); H04N (25/42); H10F (39/8037)

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a photoelectric conversion apparatus and an equipment.

Description of the Related Art

(2) Japanese Patent Laid-Open No. 2019-213159 proposes an image capturing apparatus in which two signal lines are arranged for one pixel column. A plurality of pixel circuits included in one pixel column alternately include a pixel circuit connected to one of the two signal lines and a pixel circuit connected to the other of the two signal lines. With this arrangement, it is possible to simultaneously read out signals from two pixel circuits included in the same pixel column.

SUMMARY OF THE INVENTION

(3) According to an embodiment, a photoelectric conversion apparatus comprises: a plurality of pixel circuits; a plurality of signal lines configured to read out signals from the plurality of pixel circuits; a plurality of current sources configured to supply currents to the plurality of signal lines; a switching circuit configured to switch a connection state among the plurality of signal lines and the plurality of current sources; and a control circuit configured to control the connection state of the switching circuit. The plurality of signal lines include a first signal line and a second signal line. The plurality of current sources include a first current source and a second current source. The switching circuit can implement a first connection state in which the first signal line and the second signal line are insulated from each other, the first current source is electrically connected to the first signal line, and the second current source is electrically connected to the second signal line, and a second connection state in which at least the first current source is electrically connected to at least the second signal line. The control circuit selects the first connection state in a first operation mode, and selects the second connection state in a second operation mode.

(4) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram for explaining an example of the arrangement of a photoelectric conversion apparatus according to some embodiments;
- (2) FIG. 2 is a circuit diagram for explaining an example of the arrangement of a pixel circuit according to some embodiments;
- (3) FIGS. 3A to 3G are circuit diagrams for explaining examples of the connection state of a switching circuit according to some embodiments;
- (4) FIGS. 4A to 4E are circuit diagrams for explaining examples of the circuit arrangement of the switching circuit according to some embodiments;
- (5) FIG. 5 is a timing chart for explaining an example of an operation in a high-speed operation mode according to some embodiments;
- (6) FIGS. 6A to 6D are timing charts for explaining examples of an operation in a normal operation mode according to some embodiments;
- (7) FIGS. 7A and 7B are a circuit diagram and a timing chart for explaining a modification of the photoelectric conversion apparatus according to some embodiments;
- (8) FIG. 8 is a circuit diagram for explaining a modification of the photoelectric conversion apparatus according to some embodiments;
- (9) FIG. 9 is a block diagram for explaining an example of the arrangement of a photoelectric conversion apparatus according to some embodiments;
- (10) FIGS. 10A to 10C are timing charts for explaining examples of the operation of the photoelectric conversion apparatus according to some embodiments; and
- (11) FIGS. 11A to 11C are views for explaining an equipment according to various embodiments.

DESCRIPTION OF THE EMBODIMENTS

(12) Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate.

Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

(13) Some embodiments of the present invention relate to a photoelectric conversion apparatus. Embodiments in which a photoelectric conversion apparatus is used for image capturing will mainly be described below. In this case, the photoelectric conversion apparatus can be used as an image sensor for generating an image. Furthermore, other examples of the photoelectric conversion apparatus are a distance measuring element (a sensor used for distance measurement using focus detection or Time Of Flight (TOF)), a photometric element (a sensor used to measure an incident light amount or the like), and a Light Detection and Ranging (LiDAR) sensor. Embodiments to be described below are applicable to any photoelectric conversion apparatus.

(14) An example of the arrangement of a photoelectric conversion apparatus **100** according to some embodiments will be described with reference to a block diagram shown in FIG. 1. In the example shown in FIG. 1, the photoelectric conversion apparatus **100** includes a pixel unit **110**, an A/D conversion unit **130**, a horizontal scanning circuit **140**, a signal processing circuit **150**, an output circuit **160**, a vertical scanning circuit **170**, and a timing control circuit **180**. In addition, the photoelectric conversion apparatus **100** includes switching circuits **120**, current sources **121**, and a control circuit **122**.

(15) The pixel unit **110** includes a plurality of pixel circuits **111** arranged in a two-dimensional array to form a plurality of pixel rows and a plurality of pixel columns, a plurality of driving lines **112** each extending in the direction of the pixel row, and a plurality of signal lines **113** each

extending in the direction of the pixel column. The pixel row indicates the plurality of pixel circuits **111** arranged in the horizontal direction in FIG. **1**. The pixel column indicates the plurality of pixel circuits **111** arranged in the vertical direction in FIG. **1**.

(16) One driving line **112** is arranged for each of the plurality of pixel rows. The plurality of pixel circuits **111** included in one pixel row are connected to the same driving line **112**. The two signal lines **113** are arranged for each (one) of the plurality of pixel columns. The plurality of pixel circuits **111** included in one pixel column are connected to one of the two signal lines **113** arranged for the one pixel column. In the example shown in FIG. **1**, among the plurality of pixel circuits **111** included in one pixel column, the pixel circuits **111** included in odd-numbered pixel rows from one end (for example, the upper end) of the pixel unit **110** are connected to one (for example, a signal line **113a** in FIGS. **3A** to **3G**) of the two signal lines **113**. Among the plurality of pixel circuits **111** included in one pixel column, the pixel circuits **111** included in even-numbered pixel rows from one end (for example, the upper end) of the pixel unit **110** are connected to the other (for example, a signal line **113b** in FIGS. **3A** to **3G**) of the two signal lines **113**. In this way, the plurality of pixel circuits **111** included in one pixel column are alternately connected to the different signal lines **113**. Each signal line **113** is used to read out a signal from each pixel circuit **111**.

(17) The pixel circuit **111** converts incident light into an electrical signal. Therefore, the pixel circuit **111** can also be called a light receiving circuit. The vertical scanning circuit **170** supplies a control signal to each of the plurality of pixel circuits **111** via a corresponding one of the driving lines **112**. The electrical signal is read out, via a corresponding one of the signal lines **113**, from the pixel circuit **111** supplied with the control signal of an active level.

(18) The two signal lines **113** arranged for one pixel column are connected to the same switching circuit **120**. Therefore, the photoelectric conversion apparatus **100** includes one switching circuit **120** for every two signal lines **113**. The two current sources **121** are also connected to each of the plurality of switching circuits **120**. One end of each current source **121** is connected to the switching circuit **120**, and the other end of each current source **121** is connected to a ground line. Each switching circuit **120** can switch the connection relationship among the two signal lines **113** connected to itself and the two current sources **121** connected to itself. The connection states that can be implemented by the switching circuits **120** will be described later. The control circuit **122** controls the connection state of each switching circuit **120**. For example, the control circuit **122** selects one of the plurality of connection states that can be implemented by the switching circuits **120**, and controls the switching circuits **120** to be set in the selected connection state.

(19) The A/D conversion unit **130** includes a plurality of A/D converters **131** provided for the signal lines **113**, respectively, and a control circuit **132** for controlling the operations of the A/D converters **131**. Each A/D converter **131** converts, into a digital signal, the analog signal supplied from the pixel circuit **111** via the signal line **113**. In the following description, each A/D converter **131** may be a $\Delta\Sigma$ A/D converter, a slope A/D converter, or a successive approximation type A/D converter.

(20) The horizontal scanning circuit **140** sequentially reads out the digital signals from the plurality of A/D converters **131**, and supplies them to the signal processing circuit **150**. The photoelectric conversion apparatus **100** may include, between the A/D conversion unit **130** and the horizontal scanning circuit **140**, a memory circuit for storing the digital signals. The signal processing circuit **150** processes the digital signal corresponding to the electrical signal generated in each pixel circuit **111**. The signal processing circuit **150** may perform, for example, correction processing, complement processing, or the like for the digital signal.

(21) The digital signal processed by the signal processing circuit **150** is output from the output circuit **160** to an external apparatus **190** outside the photoelectric conversion apparatus **100**. The external apparatus **190** may be, for example, a control apparatus of an equipment incorporating the photoelectric conversion apparatus **100**. The timing control circuit **180** controls the overall operation of the photoelectric conversion apparatus **100** by supplying a control signal to each

circuit of the photoelectric conversion apparatus **100**.

(22) A detailed example of the circuit arrangement of the pixel circuit **111** will be described with reference to a circuit diagram shown in FIG. **2**. The pixel circuit **111** can have an arbitrary arrangement for converting incident light into an electrical signal. An example will be described with reference to FIG. **2** but the pixel circuit **111** may have another arrangement.

(23) In the example described with reference to FIG. **2**, the pixel circuit **111** includes a photoelectric conversion element **201**, a transfer transistor **202**, a reset transistor **203**, an amplification transistor **204**, and a selection transistor **205**. In the example shown in FIG. **2**, the pixel unit **110** includes the three driving lines **112** for each pixel row, which supply control signals PRES, PTX, and PSEL to the pixel circuit **111**, respectively.

(24) The photoelectric conversion element **201** converts incident light into charges, and accumulates them. The photoelectric conversion element **201** may be, for example, a photodiode or a phototransistor. The photoelectric conversion element **201** may be a Complementary Metal Oxide Semiconductor (CMOS) sensor. Alternatively, the photoelectric conversion element **201** may be a Single Photon Avalanche Diode (SPAD) sensor.

(25) The photoelectric conversion element **201** is connected to a floating diffusion **206** via the transfer transistor **202**. The vertical scanning circuit **170** supplies the control signal PTX to the gate of the transfer transistor **202**. When the control signal PTX is set to the active level, the transfer transistor **202** is set in a conductive state. The signal charges accumulated in the photoelectric conversion element **201** are accordingly transferred to the floating diffusion **206**.

(26) The floating diffusion **206** is also connected to the gate of the amplification transistor **204**. One main electrode of the amplification transistor **204** is connected to the signal line **113** via the selection transistor **205**. The other main electrode of the amplification transistor **204** is connected to a power supply potential VDD. The vertical scanning circuit **170** supplies the control signal PSEL to the gate of the selection transistor **205**. When the control signal PSEL is set to the active level, the selection transistor **205** is set in the conductive state. The one main electrode of the amplification transistor **204** is accordingly connected to the current source **121**. This causes the amplification transistor **204** to operate as a source follower, and a signal corresponding to the potential of the floating diffusion **206** is read out into the signal line **113**.

(27) The reset transistor **203** is connected between the floating diffusion **206** and the power supply potential VDD. The vertical scanning circuit **170** supplies the control signal PRES to the gate of the reset transistor **203**. When the control signal PRES is set to the active level, the reset transistor **203** is set in the conductive state. The potential of the floating diffusion **206** is accordingly reset to the power supply potential VDD.

(28) FIGS. **3A** to **3G** are circuit diagrams for explaining various connection states among the two signal lines **113** and the two current sources **121**, which can be implemented by the switching circuit **120**. In the following description, the connection states shown in FIGS. **3A** to **3G** are represented as connection states A to G, respectively. The switching circuit **120** may be able to implement all the connection states shown in FIGS. **3A** to **3G** or only the connection states used in the respective embodiments. In FIGS. **3A** to **3G**, the two signal lines **113** connected to each switching circuit **120** are represented as the signal lines **113a** and **113b**. Furthermore, the two current sources **121** connected to each switching circuit **120** are represented as current sources **121a** and **121b**. The signal line **113a** is connected to, for example, the pixel circuits **111** included in the odd-numbered pixel rows from one end (for example, the upper end) of the pixel unit **110** among the plurality of pixel circuits **111** included in one pixel column. The signal line **113b** is connected to the pixel circuits **111** included in the even-numbered pixel rows from one end (for example, the upper end) of the pixel unit **110** among the plurality of pixel circuits **111** included in one pixel column.

(29) FIG. **3A** shows connection state A. In connection state A, the current source **121a** is electrically connected to the signal line **113a** and is insulated from the signal line **113b**. In

connection state A, the current source **121b** is insulated from the signal line **113a** and is electrically connected to the signal line **113b**. In connection state A, the signal lines **113a** and **113b** are insulated from each other.

(30) FIG. 3B shows connection state B. In connection state B, the current source **121a** is electrically connected to the signal lines **113a** and **113b**. In connection state B, the current source **121b** is insulated from the signal lines **113a** and **113b**. In connection state B, the signal lines **113a** and **113b** are electrically connected to each other.

(31) FIG. 3C shows connection state C. In connection state C, the current source **121a** is electrically connected to the signal lines **113a** and **113b**. In connection state C, the current source **121b** is electrically connected to the signal lines **113a** and **113b**. In connection state C, the signal lines **113a** and **113b** are electrically connected to each other.

(32) FIG. 3D shows connection state D. In connection state D, the current source **121a** is electrically connected to the signal line **113a** and is insulated from the signal line **113b**. In connection state D, the current source **121b** is insulated from the signal lines **113a** and **113b**. In connection state D, the signal lines **113a** and **113b** are insulated from each other.

(33) FIG. 3E shows connection state E. In connection state E, the current source **121a** is insulated from the signal line **113a** and is electrically connected to the signal line **113b**. In connection state E, the current source **121b** is insulated from the signal lines **113a** and **113b**. In connection state E, the signal lines **113a** and **113b** are insulated from each other.

(34) FIG. 3F shows connection state F. In connection state F, the current source **121a** is electrically connected to the signal line **113a** and is insulated from the signal line **113b**. In connection state F, the current source **121b** is electrically connected to the signal line **113a** and is insulated from the signal line **113b**. In connection state F, the signal lines **113a** and **113b** are insulated from each other.

(35) FIG. 3G shows connection state G. In connection state G, the current source **121a** is insulated from the signal line **113a** and is electrically connected to the signal line **113b**. In connection state G, the current source **121b** is insulated from the signal line **113a** and is electrically connected to the signal line **113b**. In connection state G, the signal lines **113a** and **113b** are insulated from each other.

(36) FIGS. 4A to 4E are circuit diagrams each for explaining the detailed circuit arrangement of the switching circuit **120** for implementing at least some of the various connection states described with reference to FIGS. 3A to 3G. The circuit arrangements for implementing the connection states shown in FIGS. 3A to 3G are not limited to the examples shown in FIGS. 4A to 4E.

(37) FIG. 4A shows the detailed circuit arrangement of the switching circuit **120** that can implement connection states A and C. The switching circuit **120** includes a switch element **401** on a path between a node **420** on a path between the signal line **113a** and the current source **121a** and a node **421** on a path between the signal line **113b** and the current source **121b**. The nodes **420** and **421** are connected to each other by the switch element **401**. The switching circuit **120** includes no circuit element on the path between the signal line **113a** and the current source **121a**, and includes no circuit element on the path between the signal line **113b** and the current source **121b**.

(38) ON/OFF of the switch element **401** is switched by a control signal supplied from the control circuit **122**. If the switch element **401** is OFF, the switching circuit **120** is in connection state A. If the switch element **401** is ON, the switching circuit **120** is in connection state C.

(39) Each of the current sources **121a** and **121b** is formed by a transistor **410**. A bias voltage V_b is supplied to the gate of the transistor **410** forming the current source **121a** and the gate of the transistor **410** forming the current source **121b**. Each transistor **410** generates a current corresponding to the bias current supplied to the gate. If the transistor **410** is electrically connected to any one of the signal lines **113**, the current is supplied to the signal line **113**.

(40) FIG. 4B shows the detailed circuit arrangement of the switching circuit **120** that can implement connection states A, B, and C. In addition to the above-described switch element **401**, the switching circuit **120** includes a switch element **402** on a path between the node **421** and the

current source **121b**. The node **421** and the current source **121b** are connected to each other by the switch element **402**. The switching circuit **120** includes no circuit element on the path between the signal line **113a** and the current source **121a**, and includes no circuit element on a path between the signal line **113b** and the switch element **402**.

(41) ON/OFF of the switch element **402** is switched by a control signal supplied from the control circuit **122**. If the switch element **401** is OFF and the switch element **402** is ON, the switching circuit **120** is in connection state A. If the switch element **401** is ON and the switch element **402** is OFF, the switching circuit **120** is in connection state B. If the switch elements **401** and **402** are ON, the switching circuit **120** is in connection state C.

(42) FIG. **4C** shows the detailed circuit arrangement of the switching circuit **120** that can implement connection states A, B, C, D, E, and G. In addition to the above-described switch elements **401** and **402**, the switching circuit **120** includes a switch element **403** on a path between the node **420** and the signal line **113a**. The node **420** and the signal line **113a** are connected to each other by the switch element **403**. The switching circuit **120** includes no circuit element on a path between the switch element **403** and the current source **121a**, and includes no circuit element on the path between the signal line **113b** and the switch element **402**.

(43) ON/OFF of the switch element **403** is switched by a control signal supplied from the control circuit **122**. If the switch element **401** is OFF, the switch element **402** is ON, and the switch element **403** is ON, the switching circuit **120** is in connection state A. If the switch element **401** is ON, the switch element **402** is OFF, and the switch element **403** is ON, the switching circuit **120** is in connection state B. If the switch elements **401**, **402**, and **403** are ON, the switching circuit **120** is in connection state C. If the switch element **401** is OFF, the switch element **402** is OFF, and the switch element **403** is ON, the switching circuit **120** is in connection state D. If the switch element **401** is ON, the switch element **402** is OFF, and the switch element **403** is OFF, the switching circuit **120** is in connection state E. If the switch element **401** is ON, the switch element **402** is ON, and the switch element **403** is OFF, the switching circuit **120** is in connection state G.

(44) FIG. **4D** shows the detailed circuit arrangement of the switching circuit **120** that can implement connection states A, C, D, F, and G. In addition to the above-described switch elements **401** and **403**, the switching circuit **120** includes a switch element **404** on a path between the node **421** and the signal line **113b**. The node **421** and the signal line **113b** are connected to each other by the switch element **404**. The switching circuit **120** includes no circuit element on the path between the switch element **403** and the current source **121a**, and includes no circuit element on a path between the switch element **404** and the current source **121b**.

(45) ON/OFF of the switch element **404** is switched by a control signal supplied from the control circuit **122**. If the switch element **401** is OFF, the switch element **403** is ON, and the switch element **404** is ON, the switching circuit **120** is in connection state A. If the switch elements **401**, **403**, and **404** are ON, the switching circuit **120** is in connection state C. If the switch element **401** is OFF, the switch element **403** is ON, and the switch element **404** is OFF, the switching circuit **120** is in connection state D. If the switch element **401** is ON, the switch element **403** is ON, and the switch element **404** is OFF, the switching circuit **120** is in connection state F. If the switch element **401** is ON, the switch element **403** is OFF, and the switch element **404** is ON, the switching circuit **120** is in connection state G.

(46) FIG. **4E** shows the detailed circuit arrangement of the switching circuit **120** that can implement all connection states A to G. In addition to the above-described switch elements **401** to **404**, the switching circuit **120** includes a switch element **405** on a path between the node **420** and the current source **121a**. The node **420** and the current source **121a** are connected to each other by the switch element **405**.

(47) ON/OFF of the switch element **405** is switched by a control signal supplied from the control circuit **122**. If the switch element **401** is OFF, the switch element **402** is ON, the switch element **403** is ON, the switch element **404** is ON, and the switch element **405** is ON, the switching circuit

120 is in connection state A. If the switch element **401** is ON, the switch element **402** is OFF, the switch element **403** is ON, the switch element **404** is ON, and the switch element **405** is ON, the switching circuit **120** is in connection state B. If the switch elements **401**, **402**, **403**, **404**, and **405** are ON, the switching circuit **120** is in connection state C. If the switch element **401** is OFF, the switch element **402** is OFF, the switch element **403** is ON, the switch element **404** is OFF, and the switch element **405** is ON, the switching circuit **120** is in connection state D. If the switch element **401** is ON, the switch element **402** is OFF, the switch element **403** is OFF, the switch element **404** is ON, and the switch element **405** is ON, the switching circuit **120** is in connection state E. If the switch element **401** is ON, the switch element **402** is ON, the switch element **403** is ON, the switch element **404** is OFF, and the switch element **405** is ON, the switching circuit **120** is in connection state F. If the switch element **401** is ON, the switch element **402** is ON, the switch element **403** is OFF, the switch element **404** is ON, and the switch element **405** is ON, the switching circuit **120** is in connection state G.

(48) An example of the operation of the photoelectric conversion apparatus **100** will be described with reference to FIGS. 5 to 6D. The photoelectric conversion apparatus **100** can operate in an operation mode selected from a high-speed operation mode of scanning the plurality of pixel rows by every two rows and a normal operation mode of scanning the plurality of pixel rows one by one.

(49) An example of the operation of the photoelectric conversion apparatus **100** in the high-speed operation mode will be described with reference to FIG. 5. The operation shown in FIG. 5 is executed by the photoelectric conversion apparatus **100** in which each switching circuit **120** can implement connection state A. During the operation in the high-speed operation mode, the control circuit **122** selects connection state A, and supplies the control signal to the switch element in each switching circuit **120** so that the switching circuit **120** is set in connection state A. The timing control circuit **180** supplies the control signal of the active level to the driving lines **112** for every two pixel rows. This can read out signals from the pixel unit **110** at high speed. With reference to FIG. 5, a description will be provided by paying attention to timings at each of which the control signal of the active level is supplied to each of the driving lines **112** on the k th to $(k+2)$ th rows. The remaining pixel rows are similarly controlled. The following description assumes that the pixel circuits **111** on the k th and $(k+2)$ th rows are connected to the respective signal lines **113a** and the pixel circuits **111** on the $(k+1)$ th row are connected to the respective signal lines **113b**.

(50) At time t_1 , the timing control circuit **180** switches, from low level to high level, the control signal PSEL to be supplied to each of the driving lines **112** on the k th and $(k+1)$ th pixel rows. This causes the amplification transistors **204** of the pixel circuits **111** on the k th row to electrically be connected to the respective signal lines **113a**, respectively, and causes the amplification transistors **204** of the pixel circuits **111** on the $(k+1)$ th row to electrically be connected to the respective signal lines **113b**, respectively.

(51) The timing control circuit **180** temporarily switches, to high level, the control signal PTX to be supplied to each of the driving lines **112** on the k th and $(k+1)$ th pixel rows while the control signal PSEL is set at high level. In connection state A, each current source **121a** supplies the bias current to the corresponding pixel circuit **111** on the k th row via the corresponding signal line **113a**, and each current source **121b** supplies the bias current to the corresponding pixel circuits **111** on the $(k+1)$ th row via the corresponding signal lines **113b**. Since the signal lines **113a** and **113b** are insulated from each other, signals are simultaneously individually read out from the pixel circuits **111** on the k th and $(k+1)$ th rows.

(52) At time t_2 , the timing control circuit **180** switches, from high level to low level, the control signal PSEL to be supplied to each of the driving lines **112** on the k th and $(k+1)$ th pixel rows. This causes the amplification transistors **204** of the pixel circuits **111** on the k th row to be insulated from the signal lines **113a**, respectively, and causes the amplification transistors **204** of the pixel circuits **111** on the $(k+1)$ th row to be insulated from the signal lines **113b**, respectively.

(53) During a period from time t_3 to time t_4 , signals are simultaneously individually read out from

the pixel circuits **111** on the (k+2)th and (k+3)th rows, similar to the period from time t₁ to time t₂. (54) An example of the operation of the photoelectric conversion apparatus **100** in the normal operation mode will be described with reference to FIGS. **6A** to **6D**. FIGS. **6A** to **6D** show individual embodiments of the photoelectric conversion apparatus **100** operating in the normal operation mode. In any of the embodiments, the timing control circuit **180** supplies the control signal of the active level to the driving line **112** for every pixel row. This can read out a signal from the pixel unit **110** with low power. In the normal operation mode, signals are individually read out from each of the plurality of pixel rows. With reference to FIGS. **6A** to **6D**, a description will be provided by paying attention to timings at each of which the control signal of the active level is supplied to each of the driving lines **112** on the kth to (k+2)th pixel rows. The remaining pixel rows are similarly controlled. The following description assumes that the pixel circuits **111** on the kth and (k+2)th rows are connected to the respective signal lines **113a** and the pixel circuits **111** on the (k+1)th row are connected to the respective signal lines **113b**.

(55) The operation according to the embodiment shown in FIG. **6A** is executed by the photoelectric conversion apparatus **100** in which each switching circuit **120** can implement connection state B. During the operation in the normal operation mode, the control circuit **122** selects connection state B, and supplies the control signal to the switch element in each switching circuit **120** so that the switching circuit **120** is set in connection state B.

(56) At time t₁, the timing control circuit **180** switches, from low level to high level, the control signal PSEL to be supplied to the driving line **112** on the kth pixel row. This causes the amplification transistors **204** of the pixel circuits **111** on the kth row to electrically be connected to the signal lines **113a**, respectively.

(57) The timing control circuit **180** temporarily switches, to high level, the control signal PTX to be supplied to the driving line **112** on the kth pixel row while the control signal PSEL is set at high level. In connection state B, each current source **121a** supplies the bias current to the corresponding pixel circuit **111** on the kth row via the corresponding signal line **113a**. By using the bias current, signals are read out from the pixel circuits **111** on the kth row via the signal lines **113a**, respectively.

(58) In connection state B, since the signal lines **113a** and **113b** are electrically connected to each other, each current source **121a** also supplies the bias current to the pixel circuit **111** connected to the signal line **113b**. However, since the signal line **113b** is insulated from any of the amplification transistors **204** of the pixel circuits **111**, no signal is read out via the signal line **113b**.

(59) At time t₂, the timing control circuit **180** switches, from high level to low level, the control signal PSEL to be supplied to the driving line **112** on the kth pixel row. This causes the amplification transistors **204** of the pixel circuits **111** on the kth row to be insulated from the signal lines **113a**, respectively.

(60) At time t₃, the timing control circuit **180** switches, from low level to high level, the control signal PSEL to be supplied to the driving line **112** on the (k+1)th pixel row. This causes the amplification transistors **204** of the pixel circuits **111** on the (k+1) row to electrically be connected to the signal lines **113b**, respectively.

(61) The timing control circuit **180** temporarily switches, to high level, the control signal PTX to be supplied to the driving line **112** on the (k+1)th pixel row while the control signal PSEL is set at high level. In connection state B, each current source **121a** supplies the bias current to the corresponding pixel circuit **111** on the (k+1)th row via the corresponding signal line **113b**. By using the bias current, signals are read out from the pixel circuits **111** on the (k+1)th row via the signal lines **113b**, respectively.

(62) In connection state B, since the signal lines **113a** and **113b** are electrically connected to each other, each current source **121a** also supplies the bias current to the pixel circuit **111** connected to the signal line **113a**. However, since the signal line **113a** is insulated from any of the amplification transistors **204** of the pixel circuits **111**, no signal is read out via the signal line **113a**.

(63) At time **t4**, the timing control circuit **180** switches, from high level to low level, the control signal **PSEL** to be supplied to the driving line **112** on the (k+1)th pixel row. This causes the amplification transistors **204** of the pixel circuits **111** on the (k+1)th row to be insulated from the signal lines **113b**, respectively.

(64) During a period from time **t5** to time **t6**, signals are read out from the pixel circuits **111** on the (k+2)th row, similar to the period from time **t1** to time **t2**.

(65) The operation according to the embodiment shown in FIG. **6B** is executed by the photoelectric conversion apparatus **100** in which each switching circuit **120** can implement connection state **C**. During the operation in the normal operation mode, the control circuit **122** selects connection state **C**, and supplies the control signal to the switch element in each switching circuit **120** so that the switching circuit **120** is set in connection state **C**. The embodiment shown in FIG. **6B** is different from that shown in FIG. **6A** in that both the current sources **121a** and **121b** supply the bias currents to the signal lines **113a** and **113b**, and the remaining points may be the same. A repetitive description will be omitted.

(66) The operation according to the embodiment shown in FIG. **6C** is executed by the photoelectric conversion apparatus **100** in which each switching circuit **120** can implement connection states **D** and **E**. During the operation in the normal operation mode, the control circuit **122** alternately selects connection states **D** and **E**, and supplies the control signal to the switch element in each switching circuit **120** so that the switching circuit **120** is set in the selected connection state.

(67) At time **t1**, the timing control circuit **180** switches, from low level to high level, the control signal **PSEL** to be supplied to the driving line **112** on the kth pixel row. This causes the amplification transistors **204** of the pixel circuits **111** on the kth row to electrically be connected to the signal lines **113a**, respectively. At time **t1**, each switching circuit **120** is in connection state **D**.

(68) The timing control circuit **180** temporarily switches, to high level, the control signal **PTX** to be supplied to the driving line **112** on the kth pixel row while the control signal **PSEL** is set at high level. In connection state **D**, each current source **121a** supplies the bias current to the corresponding pixel circuit **111** on the kth row via the corresponding signal line **113a**. By using the bias current, signals are read out from the pixel circuits **111** on the kth row via the signal lines **113a**, respectively. In connection state **D**, since the signal line **113b** is insulated from the current sources **121a** and **121b**, no bias current is supplied to the pixel circuit **111** connected to the signal line **113b**.

(69) At time **t2**, the timing control circuit **180** switches, from high level to low level, the control signal **PSEL** to be supplied to the driving line **112** on the kth pixel row. This causes the amplification transistors **204** of the pixel circuits **111** on the kth row to be insulated from the signal lines **113a**, respectively.

(70) At time **t3**, the timing control circuit **180** switches each switching circuit **120** from connection state **D** to connection state **E** by controlling the control circuit **122**.

(71) At time **t4**, the timing control circuit **180** switches, from low level to high level, the control signal **PSEL** to be supplied to the driving line **112** on the (k+1)th pixel row. This causes the amplification transistors **204** of the pixel circuits **111** on the (k+1)th row to electrically be connected to the signal lines **113b**, respectively.

(72) The timing control circuit **180** temporarily switches, to high level, the control signal **PTX** to be supplied to the driving line **112** on the (k+1)th pixel row while the control signal **PSEL** is set at high level. In connection state **E**, each current source **121a** supplies the bias current to the corresponding pixel circuit **111** on the (k+1)th row via the corresponding signal line **113b**. By using the bias current, signals are read out from the pixel circuits **111** on the (k+1)th row via the signal lines **113b**, respectively. In connection state **E**, since the signal line **113a** is insulated from the current sources **121a** and **121b**, no bias current is supplied to the pixel circuit **111** connected to the signal line **113a**.

(73) At time **t5**, the timing control circuit **180** switches, from high level to low level, the control signal **PSEL** to be supplied to the driving line **112** on the (k+1)th pixel row. This causes the

amplification transistors **204** of the pixel circuits **111** on the (k+1)th row to be insulated from the signal lines **113b**, respectively.

(74) At time **t6**, the timing control circuit **180** switches each switching circuit **120** from connection state E to connection state D by controlling the control circuit **122**. During a period from time **t7** to time **t8**, signals are read out from the pixel circuits **111** on the (k+2)th row, similar to the period from time **t1** to time **t2**.

(75) The operation according to the embodiment shown in FIG. **6D** is executed by the photoelectric conversion apparatus **100** in which each switching circuit **120** can implement connection states F and G. During the operation in the normal operation mode, the control circuit **122** alternately selects connection states F and G, and supplies the control signal to the switch element in each switching circuit **120** so that the switching circuit **120** is set in the selected connection state.

(76) The embodiment shown in FIG. **6D** is different from that shown in FIG. **6C** in that connection state F is selected instead of connection state D and connection state G is selected instead of connection state E, and the remaining points may be the same. A repetitive description will be omitted. In the embodiment shown in FIG. **6D**, both the current sources **121a** and **121b** supply the bias currents to the signal line **113a** or **113b**.

(77) The effects of the above-described embodiments shown in FIGS. **6A** to **6D** will be described below. As a comparative example, a case in which supply of the bias current from the current source **121a** via the signal line **113a** and supply of the bias current from the current source **121b** via the signal line **113b** are alternately performed will be described. If a signal is read out from the pixel circuit **111** via the signal line **113a**, no signal is read out from the pixel circuit **111** via the signal line **113b**. Therefore, the potential of the signal line **113b** lowers near a ground potential by the bias current supplied from the current source **121b**. On the other hand, the potential of the signal line **113a** rises from a potential near the ground potential to a signal voltage. This change causes the drain voltage of the transistor **410** to vary, and also causes the bias voltage **Vb** to vary via the gate-drain capacitance. If the apparatus stands by until the bias voltage **Vb** becomes stable, the speed of reading out a signal from the pixel circuit **111** is decreased.

(78) On the other hand, in any of the above-described embodiments, the same current source supplies the bias current to the pixel circuits **111** on the plurality of pixel rows (that is, the pixel circuits **111** on the odd-numbered rows and the pixel circuits **111** on the even-numbered rows). More specifically, in the embodiments shown in FIGS. **6A** and **6C**, only the current source **121a** supplies the bias current. In FIGS. **6B** and **6D**, both the current sources **121a** and **121b** supply the bias currents. In the embodiments shown in FIGS. **6C** and **6D**, since the signal line **113** that is not used to read out a signal is insulated from the pixel circuits **111** and the current sources **121**, it is set in a floating state. Therefore, this signal line holds a potential before disconnection, and does not lower near the ground potential. Thus, in the above-described embodiments, the variation of the bias voltage **Vb** as in the comparative example is suppressed, and it is thus possible to suppress a decrease in speed of reading out a signal from the pixel circuit **111**.

(79) In the embodiments shown in FIGS. **6C** and **6D**, a state in which the signal lines **113a** and **113b** are insulated from each other is maintained. This can reduce the capacity load caused by connection of the signal line **113b** when reading out a signal from the pixel circuit **111** via the signal line **113a**, and thus the photoelectric conversion apparatus **100** can operate at high speed.

(80) In the embodiments shown in FIGS. **6B** and **6D**, both the current sources **121a** and **121b** supply the bias currents to the signal line **113**. Therefore, even if the bias voltage **Vb** is lowered, a current equal to that in the high-speed operation mode in which one current source **121** supplies the bias current can be supplied to the signal line **113**. Even if the potential of the signal line **113** lowers, the transistor **410** of the current source **121** can operate in a saturated region, thereby improving the dynamic range of the signal line **113**.

(81) A modification of the above-described embodiments shown in FIGS. **6C** and **6D** will be described with reference to FIGS. **7A** and **7B**. The modification will be described as a difference

from the embodiment shown in FIG. 6C but the embodiment shown in FIG. 6D can similarly be modified. As shown in FIG. 7A, in this modification, the photoelectric conversion apparatus **100** further includes a P-type transistor **701** connected to the signal line **113a** and a P-type transistor **702** connected to the signal line **113b**.

(82) One main electrode of the P-type transistor **701** is connected to the signal line **113a**, and the other main electrode of the P-type transistor **701** is connected to a power supply voltage line. If a control signal PCL1 to be supplied to the gate of the P-type transistor **701** is at high level, the P-type transistor **701** is set in a non-conductive state. On the other hand, if the control signal PCL1 is at low level, the P-type transistor **701** is set in a conductive state, and the signal line **113a** is electrically connected to the power supply voltage line. In this way, the P-type transistor **701** functions as a stabilization circuit for stabilizing the potential of the signal line **113a**. More specifically, the P-type transistor **701** operates to stabilize the potential of the signal line **113a** when the control signal of low level is supplied, and does not perform such operation when the control signal of high level is supplied. The P-type transistor **702** also has the same arrangement as that of the P-type transistor **701**, and functions as a stabilization circuit for stabilizing the potential of the signal line **113b**.

(83) An operation according to the modification shown in FIG. 7A will be described with reference to FIG. 7B. Operations of components other than the P-type transistors **701** and **702** may be the same as those described with reference to FIG. 6C and a description thereof will be omitted.

(84) When selecting connection state D as the connection state of the switching circuit **120**, the timing control circuit **180** does not operate the P-type transistor **701** and operates the P-type transistor **702**. This fixes the signal line **113b** to the power supply voltage without being floated. When selecting connection state E as the connection state of the switching circuit **120**, the timing control circuit **180** operates the P-type transistor **701** and does not operate the P-type transistor **702**. This fixes the signal line **113a** to the power supply voltage without being floated. By stabilizing (more specifically, fixing to the power supply voltage) the potential of the signal line **113** insulating from the current source **121**, it is possible to reduce noise generating in the signal line **113** from which a signal is read out.

(85) A modification of the embodiments shown in FIGS. 6A to 6D will be described with reference to FIG. 8. A case in which the switching circuit **120** has the circuit arrangement shown in FIG. 4A will be described below but the switching circuit **120** may have another circuit arrangement.

(86) The photoelectric conversion apparatus **100** further includes a transistor **801** on a path between the current source **121a** and the switching circuit **120** and a transistor **802** on a path between the current source **121b** and the switching circuit **120**. A bias voltage V_c is supplied to the gates of the transistors **801** and **802**. The transistors **801** and **802** function as cascade transistors. In this embodiment as well, in the normal operation mode, the transistors **801** and **802** used to supply the bias currents to each of the plurality of pixel rows can be fixed. Therefore, similar to FIGS. 6A to 6D, it is possible to suppress the variation of the bias voltage V_c .

(87) The photoelectric conversion apparatus **100** further includes a sample-and-hold circuit formed by a capacitance **803** and a switch element **805** and a sample-and-hold circuit formed by a capacitance **804** and a switch element **806**. The sample-and-hold circuit formed by the capacitance **803** and the switch element **805** is connected to the current source **121a** (more specifically, the gate of the transistor **410**), and holds the bias voltage V_b to be supplied to the current source **121a**. The sample-and-hold circuit formed by the capacitance **804** and the switch element **806** is connected to the current source **121b** (more specifically, the gate of the transistor **410**), and holds the bias voltage V_b to be supplied to the current source **121b**.

(88) In the above-described photoelectric conversion apparatus **100**, the two signal lines **113** are arranged for one pixel column. Instead, three or more signal lines **113** may be arranged for one pixel column.

(89) In this case, the three or more signal lines **113** arranged for one pixel column are connected to

the switching circuit **120**. Accordingly, the current sources **121** the number of which is equal to the number of signal lines **113** are connected to the switching circuit **120**. It is possible to simultaneously read out signals from the pixel circuits **111** on rows the number of which is equal to the number of signal lines **113** arranged for one pixel column.

(90) A photoelectric conversion apparatus **900** according to another embodiment will be described with reference to FIGS. **9** to **10C**. In the photoelectric conversion apparatus **900**, one signal line **113** is arranged for each (one) of a plurality of pixel columns. A plurality of pixel circuits **111** included in one pixel column are connected to the one signal line **113** arranged for the one pixel column. Similar to the photoelectric conversion apparatus **100**, in the photoelectric conversion apparatus **900** as well, the two signal lines **113** are connected to one switching circuit **120**. Unlike the photoelectric conversion apparatus **100**, in the photoelectric conversion apparatus **900**, the two signal lines **113** connected to the one switching circuit **120** are arranged for different pixel columns.

(91) An example of the operation of the photoelectric conversion apparatus **900** will be described with reference to FIGS. **10A** to **10C**. The photoelectric conversion apparatus **900** can operate in an operation mode selected from a high-resolution operation mode of reading out a signal from each of the plurality of pixel circuits **111** and a low-resolution operation mode of reading out a signal with a resolution lower than that in the high-resolution operation mode.

(92) An example of the operation of the photoelectric conversion apparatus **900** in the high-resolution operation mode will be described with reference to FIG. **10A**. The operation shown in FIG. **10A** is executed by the photoelectric conversion apparatus **900** in which the switching circuits **120** can implement connection state A. During the operation in the high-resolution operation mode, a control circuit **122** selects connection state A, and supplies a control signal to a switch element in each switching circuit **120** so that the switching circuit **120** is set in connection state A. A timing control circuit **180** supplies a control signal of an active level to a driving line **112** for each pixel row. This can read out a signal from a pixel unit **110** with a high resolution. With reference to FIG. **10A**, a description will be provided by paying attention to timings at each of which the control signal of the active level is supplied to each of the driving lines **112** on the k th to $(k+2)$ th rows. The remaining pixel rows are similarly controlled.

(93) At time t_1 , the timing control circuit **180** switches, from low level to high level, a control signal PSEL to be supplied to the driving line **112** on the k th pixel row. This causes amplification transistors **204** of the pixel circuits **111** on the k th row to electrically be connected to the signal lines **113**, respectively.

(94) The timing control circuit **180** temporarily switches, to high level, a control signal PTX to be supplied to the driving line **112** on the k th pixel row while the control signal PSEL is set at high level. In connection state A, each current source **121** supplies a bias current to a corresponding one of the plurality of pixel circuits **111** included in the k th pixel row via a corresponding one of the signal lines **113** arranged for the respective pixel columns. Since the plurality of signal lines **113** are insulated from each other, signals are individually read out from the plurality of pixel circuits **111** included in the k th pixel row.

(95) At time t_2 , the timing control circuit **180** switches, from high level to low level, the control signal PSEL to be supplied to the driving line **112** on the k th pixel row. This causes the amplification transistors **204** of the pixel circuits **111** on the k th row to be insulated from the signal lines **113**, respectively.

(96) During a period from time t_3 to time t_4 , signals are read out from the pixel circuits **111** on the $(k+1)$ th row, similar to the period from time t_1 to time t_2 . During a period from time t_5 to time t_6 , signals are read out from the pixel circuits **111** on the $(k+2)$ th row, similar to the period from time t_1 to time t_2 .

(97) An example of the operation of the photoelectric conversion apparatus **900** in the low-resolution operation mode will be described with reference to FIG. **10B**. The operation shown in FIG. **10B** is executed by the photoelectric conversion apparatus **900** in which the switching circuits

120 can implement connection state B. During the operation in the low-resolution operation mode, the control circuit **122** selects connection state B, and supplies the control signal to the switch element in each switching circuit **120** so that the switching circuit **120** is set in connection state B. The control circuit **122** may select connection state C instead of connection state B.

(98) In the example of the operation, one signal is read out from the four pixel circuits **111** of 2 rows×2 column. The timing control circuit **180** supplies the control signal of the active level to the driving lines **112** for every two pixel rows. With reference to FIG. **10B**, a description will be provided by paying attention to timings at each of which the control signal of the active level is supplied to each of the driving lines **112** on the kth to (k+2)th pixel rows. The remaining pixel rows are similarly controlled.

(99) At time t₁, the timing control circuit **180** switches, from low level to high level, the control signal PSEL to be supplied to each of the driving lines **112** on the kth and (k+1)th pixel rows. This causes the amplification transistors **204** of the pixel circuits **111** on the kth and (k+1)th rows included in the same pixel column to electrically be connected to the same signal line **113**. In connection state B, the current sources **121** supply the bias currents to the two signal lines **113** arranged for two adjacent pixel columns.

(100) The timing control circuit **180** temporarily switches, to high level, the control signal PTX to be supplied to each of the driving lines **112** on the kth and (k+1)th pixel rows while the control signal PSEL is set at high level. In connection state B, the two signal lines **113** arranged for the two adjacent pixel columns are electrically connected to each other. Therefore, signals of the four pixel circuits **111** existing on the kth and (k+1)th rows and included in the two adjacent pixel columns are combined and read out from the two signal lines **113**.

(101) At time t₂, the timing control circuit **180** switches, from high level to low level, the control signal PSEL to be supplied to each of the driving lines **112** on the kth and (k+1)th pixel rows. This causes the amplification transistors **204** of the pixel circuits **111** on the kth and (k+1)th rows to be insulated from the signal lines **113**, respectively.

(102) During a period from time t₃ to time t₄, a signal is read out from the pixel circuits **111** on the (k+1)th and (k+2)th rows, similar to the period from time t₁ to time t₂.

(103) Another example of the operation of the photoelectric conversion apparatus **900** in the low-resolution operation mode will be described with reference to FIG. **10C**. The operation shown in FIG. **10C** is executed by the photoelectric conversion apparatus **900** in which the switching circuits **120** can implement connection states D and E. During the operation in the low-resolution operation mode, the control circuit **122** alternately selects connection states D and E, and supplies the control signal to the switch element in each switching circuit **120** so that the switching circuit **120** is set in the selected connection state. The control circuit **122** may select connection states F and G instead of connection states D and E. In the operation shown in FIG. **10C**, the photoelectric conversion apparatus **900** generates a plurality of images. The photoelectric conversion apparatus **900** sets the switching circuits **120** in connection state D, and generates one image using the half (for example, odd-numbered pixel rows) of the plurality of pixel rows and the half (for example, odd-numbered pixel columns) of the plurality of pixel columns. After that, the photoelectric conversion apparatus **900** sets the switching circuits **120** in connection state E, and generates another image using the other half (for example, even-numbered rows) of the plurality of pixel rows and the other half (for example, even-numbered pixel columns) of the plurality of pixel columns.

(104) At time t₁, the timing control circuit **180** switches, from low level to high level, the control signal PSEL to be supplied to the driving line **112** on the kth pixel row. At this time, connection state D is selected. This causes the amplification transistors **204** of the half (for example, the pixel circuits **111** on the odd-numbered columns) of the plurality of pixel circuits **111** included in the kth row to electrically be connected to the signal lines **113**, respectively.

(105) The timing control circuit **180** temporarily switches, to high level, the control signal PTX to be supplied to the driving line **112** on the kth pixel row while the control signal PSEL is set at high

level. This individually reads out signals from the half (for example, the pixel circuits **111** on the odd-numbered columns) of the plurality of pixel circuits **111** included in the kth row.

(106) At time **t2**, the timing control circuit **180** switches, from high level to low level, the control signal PSEL to be supplied to the driving line **112** on the kth pixel row. This causes the amplification transistors **204** of the pixel circuits **111** on the kth row to be insulated from the signal lines **113**, respectively.

(107) During a period from time **t3** to time **t4**, signals are read out from the half (for example, the pixel circuits **111** on the odd-numbered columns) of the plurality of the pixel circuits **111** included in the (k+2)th row, similar to the period from time **t1** to time **t2**. By repeating this operation up to the last pixel row, one image is generated. After that, the control circuit **122** switches the connection state from connection state D to connection state E.

(108) At time **t5**, the timing control circuit **180** switches, from low level to high level, the control signal PSEL to be supplied to the driving line **112** on the (k+1)th pixel row. At this time, connection state E is selected. This causes the amplification transistors **204** of the other half (for example, the pixel circuits **111** on the even-numbered columns) of the plurality of pixel circuits **111** included in the (k+1)th row to electrically be connected to the signal lines **113**, respectively.

(109) The timing control circuit **180** temporarily switches, to high level, the control signal PTX to be supplied to the driving line **112** on the (k+1)th pixel row while the control signal PSEL is set at high level. This individually reads out signals from the other half (for example, the pixel circuits **111** on the even-numbered columns) of the plurality of pixel circuits **111** included in the (k+1)th row.

(110) At time **t6**, the timing control circuit **180** switches, from high level to low level, the control signal PSEL to be supplied to the driving line **112** on the (k+1)th pixel row. This causes the amplification transistors **204** of the pixel circuits **111** on the (k+1)th row to be insulated from the signal lines **113**, respectively. By repeating this operation up to the last pixel row, another image is generated. After that, to generate a further image, the control circuit **122** switches the connection state from connection state E to connection state D.

(111) The above-described effects of the photoelectric conversion apparatus **100** can be obtained even by the photoelectric conversion apparatus **900**. A modification for the photoelectric conversion apparatus **100** may be made for the photoelectric conversion apparatus **900**.

(112) In the above-described photoelectric conversion apparatus **900**, the two signal lines **113** are arranged for one switching circuit **120**. Instead, three or more signal lines **113** may be arranged for one pixel column. In this case, the three or more signal lines **113** arranged for one pixel column are connected to the switching circuit **120**. Accordingly, the current sources **121** the number of which is equal to the number of signal lines **113** are connected to the switching circuit **120**.

OTHER EMBODIMENTS

(113) An embodiment of an equipment **1100** including a semiconductor apparatus **1103** will be described in detail with reference to FIG. **11A**. The semiconductor apparatus **1103** may correspond to a photoelectric conversion apparatus according to any one of the above-described embodiments. The semiconductor apparatus **1103** may include a semiconductor device **1101** and a package **1102** accommodating the semiconductor device **1101**. The package **1102** may include a base on which the semiconductor device **1101** is fixed and a cover made of glass or the like facing the semiconductor device **1101**. The package **1102** may further include a connection member such as a bonding wire and bump for connecting a terminal of the base and a terminal (bonding pad) of the semiconductor device **1101**.

(114) The equipment **1100** may include at least one of an optical apparatus **1104**, a control apparatus **1105**, a processing apparatus **1106**, a display apparatus **1107**, a storage apparatus **1108**, and/or a mechanical apparatus **1109**. The optical apparatus **1104** is implemented by, for example, a lens, a shutter, and a mirror. The control apparatus **1105** controls the semiconductor apparatus **1103**. The control apparatus **1105** is, for example, a semiconductor apparatus such as a Field

Programmable Gate Array (FPGA) or an Application Specific Integrated Circuit (ASIC).

(115) The processing apparatus **1106** processes a signal output from the semiconductor apparatus **1103**. The processing apparatus **1106** is a semiconductor apparatus such as a Central Processing Unit (CPU) or an ASIC for forming an Analog Front End (AFE) or a Digital Front End (DFE). The display apparatus **1107** is an Electro-Luminescence (EL) display apparatus or a liquid crystal display apparatus that displays information (image) obtained by the semiconductor apparatus **1103**. The storage apparatus **1108** is a magnetic device or a semiconductor device that stores the information (image) obtained by the semiconductor apparatus **1103**. The storage apparatus **1108** is a volatile memory such as a Static Random Access Memory (SRAM) or a Dynamic Random Access Memory (DRAM) or a nonvolatile memory such as a flash memory or a hard disk drive.

(116) The mechanical apparatus **1109** includes a moving or propulsion unit such as a motor or an engine. In the equipment **1100**, the mechanical apparatus **1109** displays the signal output from the semiconductor apparatus **1103** on the display apparatus **1107** and performs external transmission by a communication apparatus (not shown) of the equipment **1100**. To do this, the equipment **1100** may further include the storage apparatus **1108** and the processing apparatus **1106** in addition to the memory circuits and arithmetic circuits included in the semiconductor apparatus **1103**. The mechanical apparatus **1109** may be controlled based on the signal output from the semiconductor apparatus **1103**.

(117) In addition, the equipment **1100** is suitable for an electronic equipment such as an information terminal (for example, a smartphone or a wearable terminal) which has a shooting function or a camera (for example, an interchangeable lens camera, a compact camera, a video camera, or a monitoring camera). The mechanical apparatus **1109** in the camera may drive the components of the optical apparatus **1104** in order to perform zooming, an in-focus operation, and a shutter operation. Alternatively, the mechanical apparatus **1109** in the camera may move the semiconductor apparatus **1103** in order to perform an anti-vibration operation.

(118) Furthermore, the equipment **1100** can be a transportation equipment such as a vehicle, a ship, or an airplane. The mechanical apparatus **1109** in a transportation equipment can be used as a moving apparatus. The equipment **1100** as a transportation equipment may be used as an equipment that transports the semiconductor apparatus **1103** or an equipment that uses a shooting function to assist and/or automate driving (steering). The processing apparatus **1106** for assisting and/or automating driving (steering) may perform, based on the information obtained by the semiconductor apparatus **1103**, processing for operating the mechanical apparatus **1109** as a moving apparatus. Alternatively, the equipment **1100** may be a medical equipment such as an endoscope, a measurement equipment such as an analysis distance measurement sensor, an analysis equipment such as an electron microscope, or an office equipment such as a copy machine.

(119) An embodiment of an image capturing system and a moving body will be described with reference to FIGS. **11B** and **11C**. FIG. **11B** shows an example of an image capturing system **1110** concerning an in-vehicle camera. The image capturing system **1110** includes a photoelectric conversion apparatus **1111**. The photoelectric conversion apparatus **1111** may be any one of the photoelectric conversion apparatuses described in the above embodiments. The image capturing system **1110** includes an image processing unit **1112** as a processing apparatus that performs image processing for a plurality of image data acquired by the photoelectric conversion apparatus **1111**. The image capturing system **1110** also includes a parallax acquisition unit **1113** as a processing apparatus that calculates a parallax (the phase difference of a parallax image) from the plurality of image data acquired by the photoelectric conversion apparatus **1111**. In addition, the image capturing system **1110** includes a distance acquisition unit **1114** as a processing apparatus that calculates the distance to a target object based on the calculated parallax, and a collision determination unit **1115** as a processing apparatus that determines based on the calculated distance whether there is a collision possibility. In this example, the parallax acquisition unit **1113** and the distance acquisition unit **1114** are examples of an information acquisition unit that acquires

information such as distance information to a target object. That is, the distance information is information about a parallax, a defocus amount, a distance to a target object, or the like. The collision determination unit **1115** may determine the collision possibility using one of these pieces of distance information. Each of the above-described various kinds of processing apparatuses may be implemented by specially designed hardware or by general-purpose hardware for performing arithmetic processing based on a software module. Alternatively, each processing apparatus may be implemented by an FPGA, an ASIC, or the like or by a combination thereof.

(120) The image capturing system **1110** is connected to a vehicle information acquisition apparatus **1116**, and can acquire vehicle information such as a vehicle speed, a yaw rate, and a steering angle. The image capturing system **1110** is connected to a control ECU **1117** as a control apparatus that outputs a control signal to generate a braking force to the vehicle based on the determination result of the collision determination unit **1115**. That is, the control ECU **1117** is an example of a moving body control unit that controls a moving body based on distance information. The image capturing system **1110** is also connected to an alarm device **1118** that generates an alarm to the driver based on the determination result of the collision determination unit **1115**. For example, if the collision possibility is high as the determination result of the collision determination unit **1115**, the control ECU **1117** performs vehicle control to avoid a collision or reduce damage by, for example, applying the brake, returning the accelerator, or suppressing the engine output. The alarm device **1118** warns the user by, for example, generating an alarm sound or the like, displaying alarm information on the screen of a car navigation system or the like, or giving a vibration to a seat belt or steering wheel.

(121) In this embodiment, the image capturing system **1110** captures the periphery, for example, the front or rear of the vehicle. FIG. **11C** shows the image capturing system **1110** in a case in which the front of the vehicle (image capturing range **1119**) is captured. The vehicle information acquisition apparatus **1116** sends an instruction to operate the image capturing system **1110** and execute image capturing.

(122) An example in which control is performed not to cause a collision against another vehicle has been described above. However, the image capturing system can also be applied to control for automated driving following another vehicle or automated driving without deviation from a lane. Furthermore, the image capturing system can be applied not only to a vehicle such as an automobile but also to, for example, a moving body (transportation equipment) such as a ship, an airplane, or an industrial robot. The moving apparatus in the moving body (transportation equipment) includes various kinds of moving units such as an engine, motor, wheels, and propellers.

(123) In addition, the image capturing system can also be applied not only to the moving body but also to an equipment that widely uses object recognition, such as Intelligent Transport Systems (ITS).

(124) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(125) This application claims the benefit of Japanese Patent Application No. 2021-185152, filed Nov. 12, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. A photoelectric conversion apparatus comprising: a plurality of pixel circuits arranged to form a plurality of pixel rows and a plurality of pixel columns; a plurality of signal lines configured to read out signals from the plurality of pixel circuits; a plurality of current sources configured to supply currents to the plurality of signal lines; a switching circuit configured to switch a connection

state among the plurality of signal lines and the plurality of current sources; and a control circuit configured to control the connection state of the switching circuit, wherein the plurality of signal lines include a first signal line configured to read out a signal from a first pixel circuit of the plurality of pixel circuits and a second signal line configured to read out a signal from a second pixel circuit included in the same pixel column as that of the first pixel circuit, the plurality of current sources include a first current source and a second current source, the switching circuit can implement a first connection state in which the first signal line and the second signal line are insulated from each other, the first current source is electrically connected to the first signal line, and the second current source is electrically connected to the second signal line, and a second connection state in which at least the first current source is electrically connected to at least the second signal line, and the control circuit selects the first connection state and reads out a signal from the first pixel circuit and a signal from the second pixel circuit in parallel in a first operation mode, and selects the second connection state and reads out a signal from the first pixel circuit and a signal from the second pixel circuit separately in a second operation mode.

2. The apparatus according to claim 1, wherein in the second connection state, the first current source is further electrically connected to the first signal line.
3. The apparatus according to claim 2, wherein in the second connection state, the second current source is insulated from the first signal line and the second signal line.
4. The apparatus according to claim 2, wherein in the second connection state, the second current source is electrically connected to the first signal line and the second signal line.
5. The apparatus according to claim 1, wherein the switching circuit can further implement a third connection state in which the first current source is electrically connected to the first signal line, and in the second operation mode, the control circuit alternately selects the second connection state and the third connection state.
6. The apparatus according to claim 5, wherein in the second connection state, the first signal line is insulated from the first current source and the second current source, and in the third connection state, the second signal line is insulated from the first current source and the second current source.
7. The apparatus according to claim 6, wherein in the second connection state, the second current source is electrically connected to the second signal line, and in the third connection state, the second current source is electrically connected to the first signal line.
8. The apparatus according to claim 6, wherein in the second connection state, the second current source is insulated from the second signal line, and in the third connection state, the second current source is insulated from the first signal line.
9. The apparatus according to claim 6, further comprising: a first stabilization circuit configured to stabilize a potential of the first signal line; and a second stabilization circuit configured to stabilize a potential of the second signal line, wherein the control circuit operates the first stabilization circuit and does not operate the second stabilization circuit in a case where the control circuit selects the second connection state, and the control circuit operates the second stabilization circuit and does not operate the first stabilization circuit in a case where the control circuit selects the third connection state.
10. The apparatus according to claim 1, further comprising: a first transistor on a path between the first current source and the switching circuit; a second transistor on a path between the second current source and the switching circuit; a first sample-and-hold circuit configured to hold a voltage to be supplied to the first current source; and a second sample-and-hold circuit configured to hold a voltage to be supplied to the second current source.
11. The apparatus according to claim 1, wherein the switching circuit includes a first switch element on a path between a first node on a path between the first signal line and the first current source and a second node on a path between the second signal line and the second current source.
12. The apparatus according to claim 11, wherein the switching circuit satisfies at least one of a condition that a second switch element is included on a path between the second node and the

second current source, a condition that the second switch element is included on the path between the second node and the second current source and a third switch element is included on a path between the first node and the first signal line, or a condition that the third switch element is included on the path between the first node and the first signal line and a fourth switch element is included on a path between the second node and the second signal line.

13. An equipment comprising: a photoelectric conversion apparatus defined in claim 1; and at least one of an optical apparatus corresponding to the photoelectric conversion apparatus, a control apparatus configured to control the photoelectric conversion apparatus, a processing apparatus configured to process a signal output from the photoelectric conversion apparatus, a display apparatus configured to display information obtained by the photoelectric conversion apparatus, a storage apparatus configured to store the information obtained by the photoelectric conversion apparatus, or a mechanical apparatus configured to operate based on the information obtained by the photoelectric conversion apparatus.
